

Software Design Description 003

Phone-Side Gaussian Reconstruction

Brief

The WHT compression pipeline (SDD501) crushes structured motion data – 325x compression at 45 dB for a deer, 3,234x for a squirrel. But the real bottleneck isn't the transform. It's the LoRa link at 683 bytes/sec (SF7). The question becomes: what's the absolute minimum you can send and still get a recognizable image on the phone?

The answer is the **4-bit DC + Bounding Box + Gaussian** mode. Instead of sending full WHT coefficients per block, you send only the DC component (average brightness delta) quantized to 4 bits, with bounding-box addressing to eliminate per-block IDs. The phone reconstructs the smooth structure using Gaussian interpolation over the sparse DC grid. The result: a deer in **~115 bytes at 6 FPS** instead of 631 bytes at 1.1 FPS.

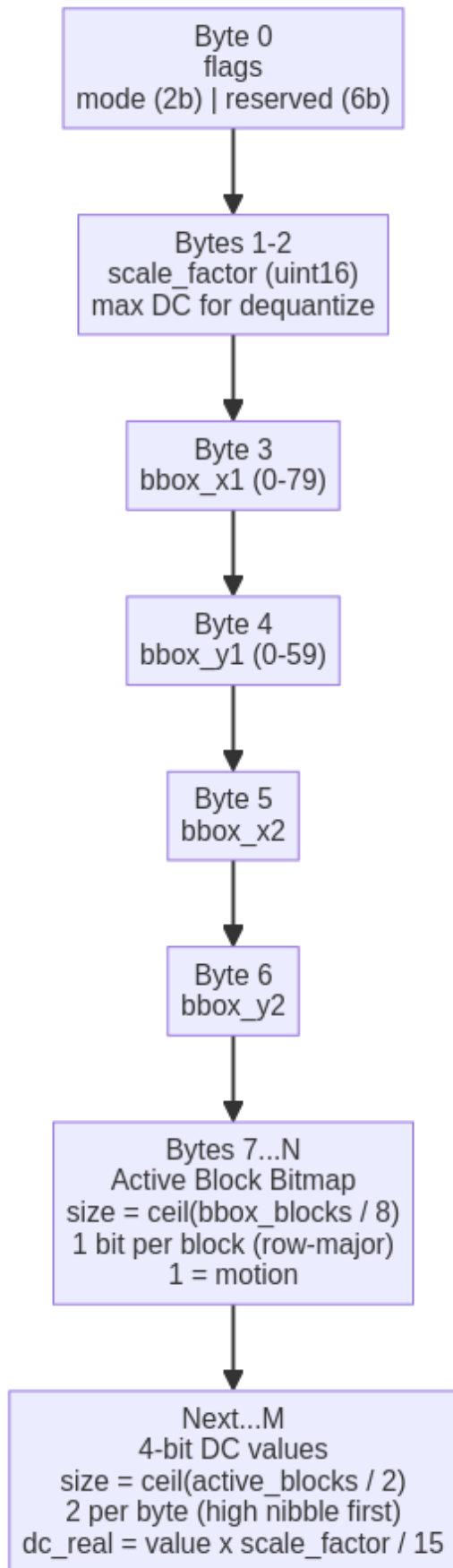
This SDD covers the receive-side reconstruction pipeline – everything that happens after the LoRa packet lands on the phone.

The Compression Stack

Each layer trades fidelity for bandwidth. They stack:

Layer 0: Raw frame	307,200 bytes	--
Layer 1: Frame differencing	only changed pixels sent	
Layer 2: Block detection (8x8, threshold)	157 blocks for a deer (~3% of frame)	
Layer 3: WHT top-N coefficient selection	325x compression at 45 dB (N=1)	
Layer 4: Packed encoding (bbox + bitmap)	487x -> 631 bytes at 1.1 FPS	
Layer 5: 4-bit DC quantization + Gaussian	2,695x -> ~115 bytes at 6.0 FPS	<- THIS SDD

Packet Format – Aggressive Mode



Byte Budget (Small Deer – 157 blocks in ~15x9 bbox)

Component	Bytes	Notes
Frame header	3	mode + scale factor
Bounding box	4	x1, y1, x2, y2
Bitmap	17	ceil(135 / 8)
DC values	79	ceil(157 / 2)
Total	103	6.6 FPS at SF7

Comparison to Other Modes

Mode	Bytes	Compression	PSNR	SF7 FPS
WHT top-1 (old encoding)	944	325x	45.0 dB	0.7
WHT top-1 (packed)	631	487x	45.0 dB	1.1
WHT top-2 (packed)	945	325x	49.4 dB	0.7
4-bit DC + Gaussian	~115	2,695x	~44.9 dB	~6.0
8-bit DC + Gaussian	~192	1,600x	~45.0 dB	3.6

Reconstruction Algorithm

The phone receives a sparse grid of 4-bit DC values positioned within a bounding box. It needs to turn this into a full-resolution diff image to overlay on the reference frame.

Step 1: Unpack & Dequantize

```

for each active block in bitmap:
    dc_4bit = next nibble from packed data
    dc_real = dc_4bit * scale_factor / 15
    diff_grid[block_y][block_x] = dc_real

```

The DC coefficient of an 8x8 WHT block is the sum of all 64 pixel values. To reconstruct block-level brightness, divide by 64 (the block area). This gives the average pixel delta for that block.

Step 2: Gaussian Interpolation (Block -> Pixel)

Each DC value represents a single brightness measurement for an 8x8 region. The raw reconstruction would be flat blocks – visible 8x8 grid artifacts. Gaussian interpolation smooths between adjacent block DC values to produce a continuous diff image.

$$\hat{d}(x, y) = \frac{\sum_{(i,j) \in \mathcal{N}} w_{ij} \cdot DC_{ij}}{\sum_{(i,j) \in \mathcal{N}} w_{ij}}$$

where the weights are:

$$w_{ij} = \exp\left(-\frac{(x - c_i)^2 + (y - c_j)^2}{2\sigma^2}\right)$$

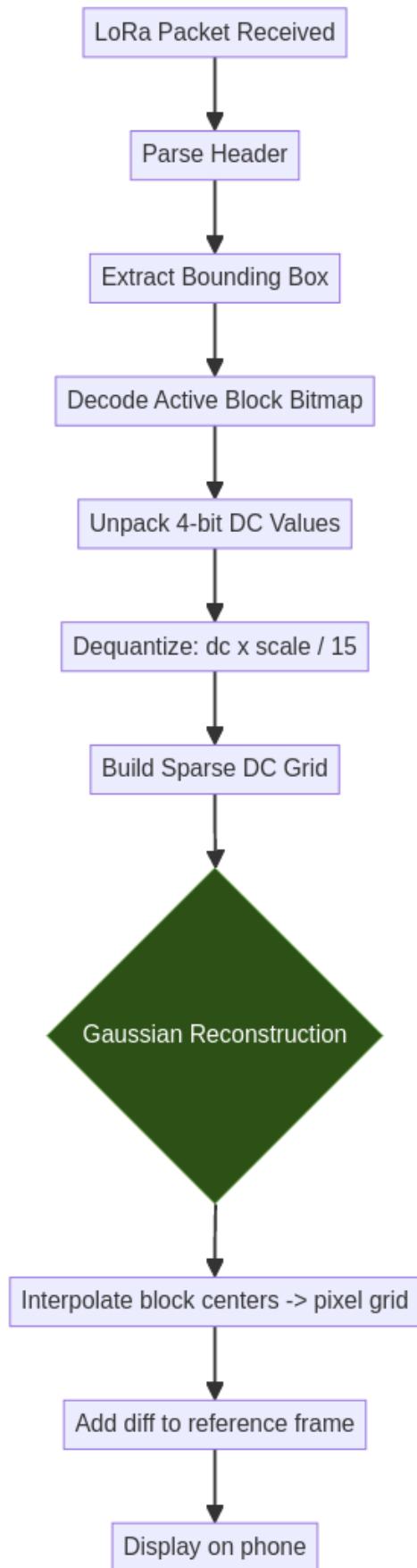
and (c_i, c_j) is the center pixel of block (i, j) , and $\mathcal{N}$ is the set of active neighboring blocks within the kernel radius.

Sigma selection: sigma = 1.0-1.5 block widths (8-12 pixels). Sigma is insensitive in this range – the sweep from sigma=0.5 to sigma=3.0 produced less than 0.04 dB variation in PSNR. The smoothing is robust because the underlying diff signal (animal silhouettes) is inherently smooth.

Step 3: Apply to Reference

```
output[y][x] = reference[y][x] + diff_reconstructed[y][x]  
clamp to [0, 255]
```

Flowchart



Implementation Notes

Why 4-bit and Not Lower?

At 4 bits (16 levels), the quantization step for a typical diff (max ~80) is about 5 intensity levels. On smooth animal silhouettes this is nearly invisible after Gaussian smoothing. At 3 bits (8 levels) the steps become 10 intensity levels – visible banding on large targets. 4 bits is the sweet spot.

Why Bounding Box + Bitmap?

Per-block addressing (2 bytes per block x 157 blocks = 314 bytes) dominates the packet at low coefficient counts. The bounding box + bitmap trick replaces 314 bytes of addressing with $4 + 17 = 21$ bytes – an **93% reduction** in overhead. This is what makes the 4-bit mode viable.

Phone Compute Budget

The Gaussian interpolation is cheap: - For a 157-block deer, you're smoothing a ~15x9 sparse grid onto a 120x72 pixel region - That's roughly 8,640 weighted averages with a 3x3 kernel - On any modern phone: <1 ms

The phone has orders of magnitude more compute than it needs. Let it do the work.

Platform

Target platform: TBD. The algorithm is pure arithmetic – no platform-specific dependencies beyond the BLE/LoRa receiver interface.

Higher FPS Ideas – Open Questions

The 4-bit DC mode gets us to ~6 FPS for a deer. To push higher:

1. **Delta-encode DC values within the bounding box.** Neighboring blocks have similar DC values. Sending deltas instead of absolutes could save 10-20 bytes, pushing closer to 7+ FPS.
2. **Adaptive quantization.** 4-bit for large smooth regions, 8-bit for edge blocks. Send a 1-bit mask for which mode each block uses.
3. **Temporal prediction.** If the animal is moving slowly, the next frame's DC values are close to the previous frame's. Send only the change-of-change. This could halve the payload for sustained motion.
4. **Variable bounding boxes.** Multiple small bounding boxes for scattered motion (3 squirrels) could be more efficient than one large sparse bbox.
5. **Run-length encoding on the bitmap.** For highly contiguous motion blobs, RLE on the active-block bitmap could save a few more bytes.
6. **LoRa SF6.** Dropping from SF7 to SF6 roughly doubles throughput to ~1,200 bytes/sec at the cost of range. At 2-mile deployment range this may still work.

Simulation Results (from SDD501 validation)

Validated on real camera frames (640x480 grayscale) with synthetic structured motion overlays:

Scenario	Changed Blocks	4-bit DC Bytes	Compression	SF7 FPS	Est. PSNR
Squirrel (~1%)	23	~20	~15,360x	~34	>45 dB
3 Squirrels (~1.5%)	70	~56	~5,486x	~12	>44 dB
Small deer (~3%)	157	~103	~2,983x	~6.6	~44.9 dB
Big deer (~17%)	864	~475	~647x	~1.4	~42 dB

Scenario	Changed Blocks	4-bit DC Bytes	Compression	SF7 FPS	Est. PSNR
Deer moved 8px	161	~106	~2,898x	~6.4	~43 dB

Related Documents

- **SDD500** – Image Processing Algorithm High-Level Description (frame acquisition, reference management, diff pipeline)
- **SDD501** – Walsh-Hadamard Transforms (the WHT math, COND_NEGATE identity, butterfly implementation, FPGA architecture)
- **SDD403** – Trail Camera IV System Architecture (ESP32-CAM -> ESP32-S3 -> FPGA -> LoRa -> Phone)